Travel With Ease

Solution Architecture, Product Scope & Implementation Roadmap

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Status: Proposed implementation baseline; client scope and legal terms subject to approval.

1. Executive summary

Build a fast, mobile-first travel agency website for a primarily Ghanaian audience. The initial release will provide packages, a personalised trip enquiry form, a package enquiry form, email notifications, WhatsApp contact, a client-editable CMS, a travel journal, and a unified world destination footprint. The solution is not a booking engine, payment platform, full CRM, or itinerary system.

The existing `faako-system` workspaces remain: `travel-with-ease-web` (Astro), `travel-with-ease-api` (Express), and `travel-with-ease-portal` (React). Reuse existing shared contracts, validation, logging and security utilities where appropriate. Preserve but defer the current estimator/currency and advanced travel features.

Architecture decisions

Layer	Selected approach	Rationale
Public website	Astro with minimal React islands	SEO, speed, low JS
Admin portal	React	Focused content and enquiry management
API	Node.js / Express	Fits existing monorepo
Database	PostgreSQL	Durable transactional storage
Public hosting	Cloudflare	CDN, TLS and edge delivery
API / database hosting	Railway	Existing operational familiarity
Media / off-site backups	Cloudflare R2	Object storage with private access
Email	Resend	Transactional notifications
Bot protection	Cloudflare Turnstile	Public form abuse controls

2. Scope and exclusions

Launch scope

- Public pages: Home, Destinations/Packages, Services, About/Testimonials, Travel Journal, FAQs, Contact and legal pages.
- Package detail pages with indicative 'from' prices, date/tier information and clear non-binding price disclosures.
- Two independent enquiry forms: personalised trip and package-specific.
- Database-backed enquiry inbox with simple New, Contacted and Closed statuses.

- Email notification queue, delivery status and retries; WhatsApp click-to-chat.
- Structured CMS for selected text, FAQs, testimonials, packages, journal, images, metadata and contact details.
- Draft, preview, publish/unpublish, revision history and basic rollback.
- Unified world map combining completed and confirmed upcoming trips into one country intensity.
- SEO/AEO, analytics with applicable consent handling, security, backups and handover.

Deferred / separately scoped

- Online payments, Paystack checkout, invoicing, live exchange rates and estimator.
- Automated WhatsApp messaging, full CRM, itinerary management and customer portal.
- Automated social synchronisation, AI-generated travel itineraries and advanced analytics.

3. Information architecture and customer journey

PROCESS FLOW

AWebsite → BHome

A → CDestinations and Packages

A → DServices and About

A → ETravel Journal

A → FFAQs and Contact

A → GLegal Pages

B → HUnified World Map

B → IPlan a Trip

C → JPackage Details

J → KPackage Enquiry

I → LPersonalised Enquiry

K → MConfirmation

L → M

The homepage should follow an editorial sequence: photographic hero and clear CTA; featured packages; travel footprint; how it works; travel journal; testimonials/FAQs; final enquiry CTA. Mobile navigation and form usability are first-class requirements.

4. Design direction and quality bar

Avoid a generic AI-generated aesthetic. Do not default to gradients, glassmorphism, repeated cards, ubiquitous eyebrows/kickers, glowing borders, decorative blobs, oversized rounded corners or excessive motion. Use a disciplined grid, intentional whitespace, real travel photography, confident typography, clear hierarchy and strong editorial composition. Each section must earn its place; avoid repeating the same card grid. Keep copy concise and specific. Use the brand palette: navy `#010c33`, cream `#f4ede3`, gold `#d8b56a`, grey `#b6b1a9`, blue `#7ab8ff`, dark `#000a1f`.

React Bits may be used selectively (e.g. Split Text, Scroll Reveal, Spotlight Card or a restrained hero background), only where it improves comprehension or craft. No gratuitous animations; support

reduced motion, keyboard access, touch devices and strong Core Web Vitals. Maintain Astro SSR/static discoverability rather than rendering core copy only in React.

5. Target architecture

PROCESS FLOW

VVisitor → WAstro Public Website

W → CFCloudflare CDN and TLS

W → APIExpress API on Railway

STAFFAuthenticated Staff → PORTALReact CMS Portal

PORTAL → API

API → DB(PostgreSQL)

API → R2Cloudflare R2 Media

API → QNotification Outbox

Q → EMAILResend

DB → BACKUPScheduled Encrypted Backup

BACKUP → PRIVATEPrivate R2 Backup Bucket

W → ANALYTICSConsent-aware Analytics

W → WAWhatsApp Click-to-Chat

PORTAL -. optional authorised import .-> METAMeta Graph APIs

The public browser can access only approved public endpoints. Staff content and enquiry operations require server-side authorization. Do not expose API secrets, private records or privileged R2 URLs to browsers. Public content publishing must trigger controlled cache invalidation/rebuild with a visible publication state and retries.

6. CMS and publishing workflow

Editable: selected hero copy, service descriptions, packages, FAQs, testimonials, journal articles, images, contact details, CTAs and SEO fields. Non-editable: application layout, navigation structure, security rules, validation copy and code-owned components.

PROCESS FLOW

AStaff Edits → BSchema and Media Validation

B → CDraft

C → DPreview

D → E{Publish?}

E → |No| C

E → |Yes| FVersioned Publication

F → GRebuild or Revalidate

G → HPublic Page Updated

F → IAudit Record

The portal navigation should be Overview, Enquiries, Website Content, Packages, Travel Journal, Travel Footprint, Media Library and Settings. Single-user launch scope is sufficient; still implement secure sessions and authoritative access checks. Sanitize rich text, restrict media type/size, and prevent stored XSS.

7. Unified world travel footprint

One map, no completed/upcoming switch. Country intensity reflects `total = completed trips + confirmed upcoming trips`. Enquiries, tentative bookings and cancelled trips do not count. A multi-country trip may count once per visited destination country; define deduplication and historical import rules.

Use accurate geographic boundaries with ISO country codes. Render a responsive accessible choropleth with a subdued base, a restrained gold intensity scale, tooltips/tap details, and an accessible country list/table fallback. Use a logarithmic or quantile scale with a documented legend; no fabricated activity numbers. Display only aggregate counts, never traveller identities.

PROCESS FLOW

AStaff Records Historical or Confirmed Trip Activity → BValidate Country, Status and Count

B → CStore Audited Activity

C → DAggregate by ISO Country

D → EPublic Map Dataset

E → FUpdate Country Fill Intensity

C → GRevision and Audit History

Proposed fields: activity ID, country ISO code, count, completed/confirmed-upcoming status, period, source, publication flag, editor and timestamps. Expose aggregate totals only. Future booking integration can replace manual entry without changing the public contract.

8. Enquiries and notifications

PROCESS FLOW

AVisitor → B{Form Type}

B → CCustom Trip

B → DPackage Enquiry

C → EValidate and Turnstile Verify

D → E

E → F{Valid?}

F → |No| GAccessible Field Errors

F → |Yes| HPersist Enquiry

H → ICreate Notification Outbox Event

H → JShow Confirmation and Reference

I → KResend with Retry

H → LProtected Staff Inbox

L → MContact and Update Status

Collect only name, email/phone, destination/package, preferred dates, traveller count, budget range where relevant, preferences and optional notes. Store source, timestamps and consent/notice evidence. Do not collect passport scans, birth dates, card data or other sensitive documents at launch. Do not make enquiry persistence dependent on email delivery. Marketing opt-in must be separate.

9. Travel Journal and social content

Journal entries should have title, slug, excerpt, rich-text body, featured image, destination tags, related package, source post URL, publication state and SEO metadata. Launch with manually linked/curated posts; automated Meta importing is optional and contingent on account eligibility, OAuth permissions and review.

PROCESS FLOW

AOriginal Social Post → BLink or Approved Import

B → CEditorial Draft

C → DReview Rights and Accuracy

D → EPublish Original Journal Content

E → FIndexed Article

F → GRelated Package CTA

Respect image/content rights; never assume a social embed provides durable searchable content.

10. Integration register

Service	Connection	Launch decision
Cloudflare Pages / DNS / TLS	Git deployment, DNS and SSL	Required
Cloudflare R2	S3-compatible API	Required
Railway / PostgreSQL	Deployment and DB credentials	Required
Resend	Email REST API	Required
Cloudflare Turnstile	Client token and server verification	Required
WhatsApp	Click-to-chat URL	Required
Google Search Console	Domain verification, sitemap	Required
Google Analytics 4	Consent-aware measurement	Required
Meta Instagram/Facebook Graph APIs	OAuth, approved content retrieval	Optional
Paystack	API and signed webhooks	Deferred
WhatsApp Business Platform	Meta Cloud API	Deferred
Exchange-rate provider	Server-side rates API	Deferred

Proposed internal API

- `GET /api/public/content/:page`, `/packages`, `/articles/:slug`, `/travel-footprint`
- `POST /api/public/inquiries` with a discriminated enquiry type.

- Authenticated `/api/agent/content/*`, `/packages/*`, `/articles/*`, `/travel-activity/*`, `/media/*`, `/inquiries/*`.
- `/api/auth/*` for reviewed staff authentication; background notification and backup jobs are not public endpoints.

11. Data protection, security and legal readiness

Target Ghana Data Protection Act 2012 (Act 843) compliance and verify Data Protection Commission registration obligations and business-specific requirements. Have the client approve its legal terms. Provide Privacy Policy, Website Terms, Booking/Payment/Cancellation Terms, Cookie Policy/Preferences, and contact/privacy request information. Booking refunds, supplier responsibility, visa guidance and cancellation rules must reflect actual approved business policy.

Security controls: least privilege, MFA where feasible, session rotation and expiry, server-side authorization, input schemas, rich-text sanitization, file scanning/restrictions, TLS, encryption at rest, secret rotation, private buckets, Turnstile and rate limiting, dependency checks, audit logs, monitoring and incident procedures. Target WCAG 2.2 AA and a risk-based OWASP ASVS Level 2 assessment.

Backups: daily PostgreSQL export to a private encrypted R2 bucket; proposed retention 7 daily, 4 weekly and 6 monthly copies, subject to cost validation. Alert on failures, test restores and document actual RPO/RTO. Daily snapshots imply an RPO up to about 24 hours, not point-in-time recovery.

12. Delivery roadmap

Week	Focus	Exit gate
1	Scope baseline, repository audit, design system, data schema, storefront and package templates	Approved designs and functioning foundation
2	Secure CMS, authentication, media, enquiry persistence and notifications	Client can edit content and receive stored enquiries
3	Journal, unified world map, social links, SEO, analytics and restrained animation	End-to-end publication and map updates
4	Legal review, security/performance/accessibility QA, backup/restore, UAT, deployment and training	Signed release checklist and production smoke tests

PROCESS FLOW

ARequirements Baseline → BDesign Approval

B → CVertical Slice Implementation

C → DAutomated Tests

D → EStaging UAT

E → F{Accepted?}

F → |No| C

F → |Yes| GProduction Readiness

G → HLaunch

H → IHypercare and Handover

Four weeks is a target, not a guarantee; Meta API permissions, legal approvals and content delays can affect timing. Maintain feature-level acceptance criteria, owners, dependencies, estimates and change controls.

13. Acceptance and quality gates

- Both enquiry forms validate, persist and return a reference; notification failures retry without losing enquiries.
- Client can draft, preview, publish, unpublish and revert approved content; SEO metadata updates on publication.
- World map shows one combined activity intensity and correct country totals; historical adjustments are audited.
- Portal blocks unauthorised access; media and secrets are protected.
- Site works on common mobile devices, with keyboard access and reduced-motion behaviour.
- Sitemap, canonical URLs, structured data and analytics configuration are verified.
- Legal pages are approved; privacy request and retention procedures exist.
- Backups restore successfully; production rollback and monitoring are documented.
- No unresolved critical/high findings without documented risk acceptance.

14. Risks, assumptions and decisions

Risk / dependency	Mitigation
Scope creep	Approved MVP backlog; add-ons via change control
Missing package details and imagery	Content owner and deadlines; approved placeholders only
Meta API permissions	Manual social links at launch
Astro content staleness	Tested publishing/rebuild mechanism
Incorrect historical map data	Explicit count rules, validation and audit
Notification outages	Outbox, retries, monitoring
Legal/privacy approval delay	Review before production
Hosting cost increases	Monitor usage and set thresholds
Compressed four-week timeline	Vertical slices and prioritised acceptance gates

Commercial baseline: GHS 4,000 development and GHS 250/month managed hosting/maintenance from the prior proposal. The CMS, journal and interactive map are material scope additions: estimate effort and confirm inclusion or a revised fee before treating the commercial baseline as final.

15. Repository deliverables and next action

Update `docs/apps/travel-with-ease/` with this scope baseline, architecture decisions, data dictionary, API contracts, security controls, integration inventory, environment matrix, deployment/backup runbooks, test cases, UAT evidence, release notes and change log.

First Codex task: audit the current `develop` branch against this plan, identify existing versus missing capabilities, produce a file-level gap matrix and sequenced implementation backlog. Do not

rewrite the application or make unrelated changes. Preserve uncommitted work and avoid touching other monorepo applications.